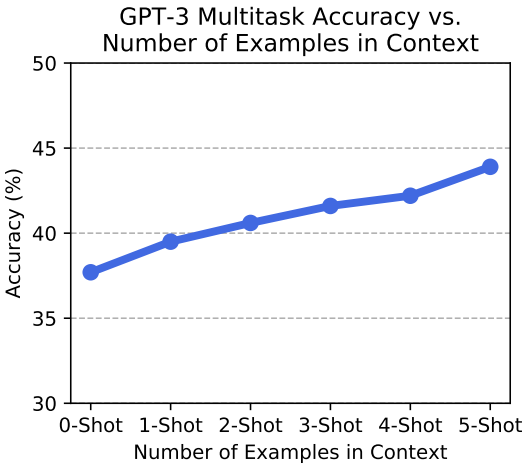


Figure 10: As the number of few-shot instruction examples increases, the accuracy monotonically increases. Notably, zero-shot performance is only somewhat lower than 5-shot accuracy.

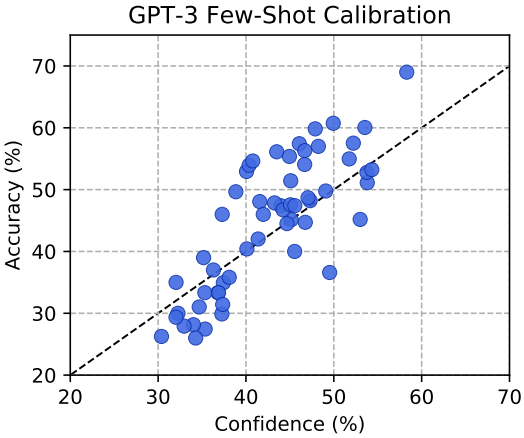
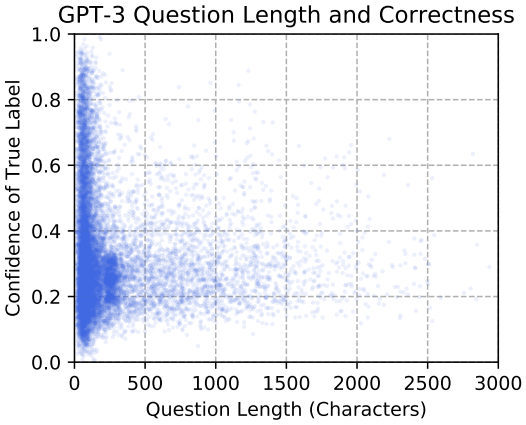


Figure 11: While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is $r = 0.81$, compared to $r = 0.63$ in the zero-shot setting.

B Test Details

B.1 Task Descriptions and Examples

We provide analysis of question length and difficulty in Figure 12. We list all tasks and the topics they test in Table 2. We also provide an example for each task starting with Figure 14.



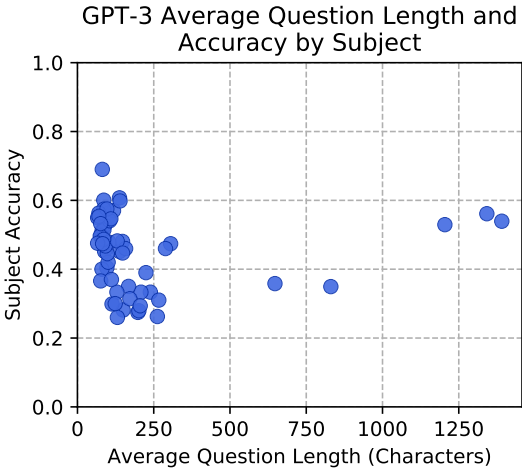


Figure 12: Figures on the relation between question difficulty and question length. For questions longer than a tweet (280 characters), the correlation between question length and true label confidence is slightly positive. This shows that longer questions are not necessarily harder.

B.2 Exact Question and Answer Contamination

Since language models train on vast text corpora, there is some chance that they have seen the exact question and answer during pretraining. If they memorized the exact question and answer, then they would attain higher accuracy than their true ability. Likewise, a question's entropy would be especially low if it were memorized. Memorized questions and answers should have low entropy and

high accuracy. However, in Figure 13, we see that accuracy and question entropy are not positively correlated, suggesting that the test's low-entropy questions do not correspond to memorized (and thereby correctly predicted) answers. This suggests that our *exact* questions were not memorized. However, during pretraining models encountered text *related* to our questions through processing Wikipedia. We also note that most of our questions came from PDFs or websites where questions and answers are on separate pages.

See Brown et al. (2020) for a previous discussion of contamination showing that the phenomena hardly affects performance. To reduce the probability that future models encounter exact questions during test-time, we will provide a list of question sources.

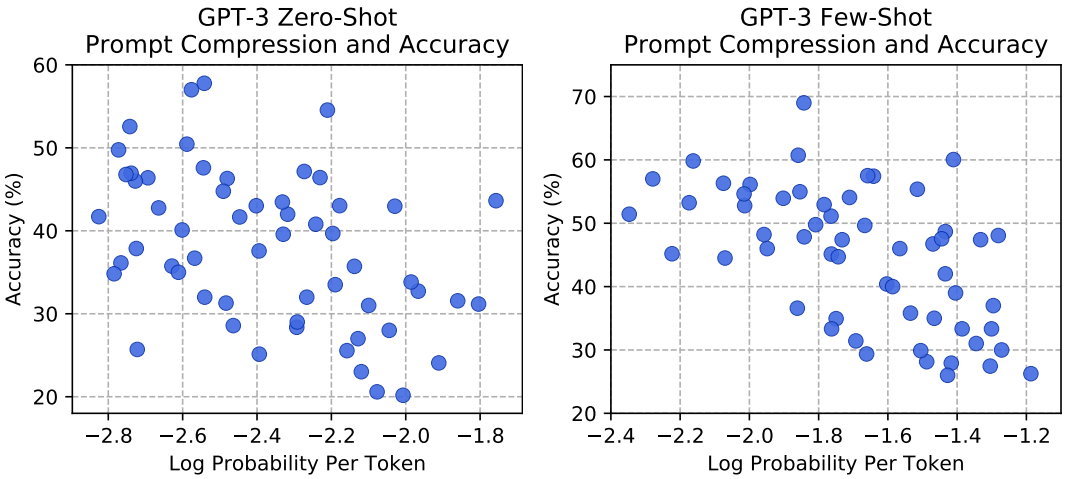


Figure 13: The average log probability of the question (without answer) is not strongly positively correlated with accuracy, all else equal. Each point corresponds to a task. Higher log probability indicates higher compression, and especially high log probability would suggest memorization. In the zero-shot question prompt, the correlation between average log probability and accuracy is $r = -0.43$, and for the few-shot setting the correlation is $r = -0.56$.

Task	Tested Concepts	Supercategory
Abstract Algebra	Groups, rings, fields, vector spaces, ...	STEM
Anatomy	Central nervous system, circulatory system, ...	STEM
Astronomy	Solar system, galaxies, asteroids, ...	STEM
Business Ethics	Corporate responsibility, stakeholders, regulation, ...	Other
Clinical Knowledge	Spot diagnosis, joints, abdominal examination, ...	Other
College Biology	Cellular structure, molecular biology, ecology, ...	STEM
College Chemistry	Analytical, organic, inorganic, physical, ...	STEM
College Computer Science	Algorithms, systems, graphs, recursion, ...	STEM
College Mathematics	Differential equations, real analysis, combinatorics, ...	STEM
College Medicine	Introductory biochemistry, sociology, reasoning, ...	Other
College Physics	Electromagnetism, thermodynamics, special relativity, ...	STEM
Computer Security	Cryptography, malware, side channels, fuzzing, ...	STEM
Conceptual Physics	Newton's laws, rotational motion, gravity, sound, ...	STEM
Econometrics	Volatility, long-run relationships, forecasting, ...	Social Sciences
Electrical Engineering	Circuits, power systems, electrical drives, ...	STEM
Elementary Mathematics	Word problems, multiplication, remainders, rounding, ...	STEM
Formal Logic	Propositions, predicate logic, first-order logic, ...	Humanities
Global Facts	Extreme poverty, literacy rates, life expectancy, ...	Other
High School Biology	Natural selection, heredity, cell cycle, Krebs cycle, ...	STEM
High School Chemistry	Chemical reactions, ions, acids and bases, ...	STEM
High School Computer Science	Arrays, conditionals, iteration, inheritance, ...	STEM
High School European History	Renaissance, reformation, industrialization, ...	Humanities
High School Geography	Population migration, rural land-use, urban processes, ...	Social Sciences
High School Gov't and Politics	Branches of government, civil liberties, political ideologies, ...	Social Sciences
High School Macroeconomics	Economic indicators, national income, international trade, ...	Social Sciences
High School Mathematics	Pre-algebra, algebra, trigonometry, calculus, ...	STEM
High School Microeconomics	Supply and demand, imperfect competition, market failure, ...	Social Sciences
High School Physics	Kinematics, energy, torque, fluid pressure, ...	STEM
High School Psychology	Behavior, personality, emotions, learning, ...	Social Sciences
High School Statistics	Random variables, sampling distributions, chi-square tests, ...	STEM
High School US History	Civil War, the Great Depression, The Great Society, ...	Humanities
High School World History	Ottoman empire, economic imperialism, World War I, ...	Humanities
Human Aging	Senescence, dementia, longevity, personality changes, ...	Other
Human Sexuality	Pregnancy, sexual differentiation, sexual orientation, ...	Social Sciences
International Law	Human rights, sovereignty, law of the sea, use of force, ...	Humanities
Jurisprudence	Natural law, classical legal positivism, legal realism, ...	Humanities
Logical Fallacies	No true Scotsman, base rate fallacy, composition fallacy, ...	Humanities
Machine Learning	SVMs, VC dimension, deep learning architectures, ...	STEM
Management	Organizing, communication, organizational structure, ...	Other
Marketing	Segmentation, pricing, market research, ...	Other
Medical Genetics	Genes and cancer, common chromosome disorders, ...	Other
Miscellaneous	Agriculture, Fermi estimation, pop culture, ...	Other
Moral Disputes	Freedom of speech, addiction, the death penalty, ...	Humanities
Moral Scenarios	Detecting physical violence, stealing, externalities, ...	Humanities
Nutrition	Metabolism, water-soluble vitamins, diabetes, ...	Other
Philosophy	Skepticism, phronesis, skepticism, Singer's Drowning Child, ...	Humanities
Prehistory	Neanderthals, Mesoamerica, extinction, stone tools, ...	Humanities
Professional Accounting	Auditing, reporting, regulation, valuation, ...	Other
Professional Law	Torts, criminal law, contracts, property, evidence, ...	Humanities
Professional Medicine	Diagnosis, pharmacotherapy, disease prevention, ...	Other
Professional Psychology	Diagnosis, biology and behavior, lifespan development, ...	Social Sciences
Public Relations	Media theory, crisis management, intelligence gathering, ...	Social Sciences
Security Studies	Environmental security, terrorism, weapons of mass destruction, ...	Social Sciences
Sociology	Socialization, cities and community, inequality and wealth, ...	Social Sciences
US Foreign Policy	Soft power, Cold War foreign policy, isolationism, ...	Social Sciences
Virology	Epidemiology, coronaviruses, retroviruses, herpesviruses, ...	Other
World Religions	Judaism, Christianity, Islam, Buddhism, Jainism, ...	Humanities

Table 2: Summary of all 57 tasks.